A QUICK REFERENCE GUIDE TO USING R FUNCTIONS

Summary

The Statistics Team members within the Surveillance Branch (SB), DHQP, have developed a series of statistical comparison functions. These functions were developed to conduct statistical tests for epidemiological measures, consistent with the methods used by NHSN.

Initial statistical comparison functions tested several common epidemiological measures such as incidence density rates, proportions, and ratios such as standardized metrics known as the standardized infection ratio (SIR), standardized utilization ratio (SUR), standardized antimicrobial administration ratio (SAAR), etc., and all statistical tests are based on the mid-p exact method.

The document also describes additional R function that help complement the statistical analysis of epidemiolocal data. This function compares summary statistics such as medians and empirical cumulative distribution functions using nonparametric methods such as the Brown-Mood median and both Anderson-Darling and Dowd DTS ECDF tests.

Overview

R functions written by NHSN statisticians are listed below:

Description	Program Filename	Function Name
1. Testing SIR=1	MID-P FOR SIR EQUAL TO 1	sir_function
2. Comparing 2 SIRs (ratio)	MID-P FOR COMPARING 2 SIRs	binom
3. 95%CI of incidence density rate (single rate)	MID-P FOR INCIDENCE DENSITY RATE	rateCIcomp
4. Comparing 2 rates (ratio)	MID-P FOR COMPARING 2 RATES	binom_rate
5. 95%CI of single proportion	MID-P FOR SINGLE PROPORTION	binom_midp_ci
6. Comparing 2 proportions (ratio)	MID-P FOR COMPARING 2 PROPORTIONS.	Pcomp

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7. Comparing two sample medians and ECDFs	Comparing Two ECDFs.R	compare_two_ecdfs

To facilitate user application, a brief description, accompanying example data and corresponding outputs are provided below for each function.

1. MID-P FOR SIR EQUAL TO 1

Description
This function can be used to test whether SIR (or similar standardized metrics like SUR, SAAR) is statistically different from 1, consistent with the methods used by NHSN.

Example dataset and Input parameters
Input parameters:
obs=observed value, exp=predicted value

```
#example data
df <- data.frame(
  id    = c(1, 2, 3),
  num   = c(1, 2, NA),
  denom = c(2, 0.6, 3.3)
)
# Apply the sir_function row-wise
df1 <- df %>%
  rowwise() %>%
  mutate(result = list(sir_function(obs=num, pred=denom)))%>%
  unnest_wider(result)

# Display the results
head(df1)
```

Outputs and interpretations

Abbreviations
Obs=observed value
Exp=predicted value
Midp=2 tailed mid-P value
LLIMIT=lower limit of 95% CI
ULIMIT=upper limit of 95% CI
Note:
*No results will be generated if entry values are missing or impossible values.

*When SIR=0, the lower limit (LLIMIT) of 95%CI is set to missing by default. A user may overwrite LLIMIT=0 if desired.

```
# A tibble: 3 x 7
     id   num denom   midp   sir  llimit ulimit
  <dbl> <dbl> <dbl>  <dbl> <dbl>   <dbl>  <dbl>
1     1     1     2  0.541   0.5  0.0250   2.47
2     2     2   0.6 NA      NA   NA      NA
3     3    NA   3.3 NA      NA   NA      NA
```

- First example (First row of output): The mid-p-value is 0.541. Because this p-value is greater than 0.05, we would conclude that computed SIR value=0.5 is no statistically different from 1.0. 95%CI also includes the null (1.0)
- Predicted value of second example is <1.0 so that SIR and 95%CI are not computed.
- Observed value of third example is missing so that SIR and 95%CI are not computed.

REFERENCE:

1. JH ABRAMSON. WINPEPI PROGRAMS. DESCRIBE MANUAL (VERSION 2.42), PAGE-52. Available at 'http://www.brixtonhealth.com/pepi4windows.html'
2. Dean AG, Sullivan KM, Soe MM. OpenEpi: Open Source Epidemiologic Statistics for Public Health, Version. www.OpenEpi.com.
3. Rothman KJ, Boice JD Jr: Epidemiologic analysis with a programmable calculator. NIH Pub No. 79-1649. Bethesda, MD: National Institutes of Health, 1979;31-32.
4. Rothman KJ, Greenland S. Modern Epidemiology, 2nd Edition. Lippincott-Raven Publishers, Philadelphia, 1998.
5. Geoffrey rc, shu-ying y. Mid-p confidence intervals for the poisson expectation. Statistics in medicine, 13,2189-2203 (1994)

2. MID-P FOR COMPARING 2 SIRs

Description

This function can be used to perform a comparison between two SIRs, consistent with the methods used by NHSN. SIR could be substituted by similar standardized metrics like SUR, SAAR.

Example dataset and input parameters

Data input layout:

O1= observed events in group1

E1= predicted events in group1

O2= observed events in group2

E2= predicted events in group2

```
#example data
df <- data.frame(
  id       = c(1,2,3,4),
  obs2017  = c(1,1,1,3),
  pred2017 = c(1,0.5,2,NA),
  obs2016  = c(2,1,0,4),
  pred2016 = c(3,2,5.2,5.2)
)

# Apply the binom function row-wise
df1 <- df %>%
  rowwise() %>%
  mutate(binom_result = list(binom(O1 = obs2017, E1 = pred2017, O2 = obs2016, E2 = pred2016))) %>%     #not a
cleaner syntax: binom(obs2017, pred2017, obs2016, pred2016)
  unnest_wider(binom_result) # %>%
 mutate(across(c(midP, RATIO, LL, UL), ~ round(., 3)))

# Display the results
head(df1)
```

Outputs and interpretations

Abbreviations

midP= 2 tailed mid-P value

RATIO= SIR in group2 / SIR in group1 = SIR2/SIR1 = (&O2/&E2)/(&O1/&E1)

LL= 95% lower bound of RATIO

UL= 95% Upper bound of RATIO

Note: When RATIO=0, the lower limit (LL) of 95%CI is set to missing by default. A user may change it to LL=0 if desired.

```
# A tibble: 4 x 9
     id obs2017 pred2017 obs2016 pred2016   midP  RATIO      LL    UL
  <dbl>   <dbl>    <dbl>   <dbl>    <dbl>  <dbl>  <dbl>   <dbl> <dbl>
1     1       1        1       2        3  0.734  0.667  0.0507  19.7
2     2       1      0.5       1        2 NA     NA     NA     NA
3     3       1        2       0      5.2  0.278  0      NA      7.31
4     4       3       NA       4      5.2 NA     NA     NA     NA
```

- First comparison (first row): SIR for 2017 is 1/1 and SIR for 2016 is 2/3. The ratio of 2 SIRs (2016 SIR/ 2017 SIR) is shown as 0.667 with 95%CI. As the midP value is >0.05, we would conclude that there is NO significant difference between the two SIRs.

- Second example shows predicted number<1 for pred2017 so that mid-p and ratio are not computed.
- Third example has ratio of 0 so that the lower limit of 95%CI is not computed.
- Fourth example has missing predicted value for pred2017 so that mid-p and ratio are not computed.

Reference

1. Breslow NE, Day NE. Statistical methods in cancer research. Volume II--The design and analysis of cohort studies. IARC Sci Publ. 1987;(82):1-406. PMID: 3329634. http://www.iarc.fr/en/publications/pdfs-online/stat/sp82/SP82_vol2-3.pdf
2. Erich L. Lehmann, Joseph P. Romano. Testing Statistical Hypotheses. Wiley, New York, 1959 (Third edition, 2005, Springer Texts in Statistics)
3. John Pezzullo. Interactive Statistical Calculation Pages. www.Statpages.com.
4. Dean AG, Sullivan KM, Soe MM. OpenEpi: Open Source Epidemiologic Statistics for Public Health, Version. www.OpenEpi.com, updated 2013/03/21

3. MID-P FOR INCIDENCE DENSITY RATE

Description

This function can be used to generate confidence interval of a single incidence density rate (e.g., CLABSI rate), consistent with methods used by NHSN.

Example dataset and input parameters

Input parameters:

numer=numerator, denom=denominator.

```
# EXAMPLE DATASET
df <- data.frame(
   id  = c(1, 2),
   obs = c(0,10),
   pt  = c(10,10)
)


# Apply the function row-wise and add multiple columns
df1 <- df %>%
  rowwise() %>%
  mutate(result = list(rateCIComp(obs,pt)))%>%  #replace with num&denom in dframe
  unnest_wider(result)
df1 <- df1 %>% mutate(across(c(rate, rate_l, rate_u), ~ format(., nsmall = 3)))#keep trailing zeros
```

```
# df1<-df1 %>% mutate(across(c(rate, rate_l, rate_u), ~ round(., 3)))    #format to 3 decimal places

# Display the results
head(df1)
```

Outputs and interpretations

The rate here is multiplied by 1,000 so that Incidence density rate is per 1,000 person-time & rate_l & rate_u represents.lower and upper limit of 95%CI respectively.
Note: When rate=0, the lower limit (rate_l) of 95%CI is set to missing by default. A user may change it to rate_l=0 if desired.

```
# A tibble: 2 x 6
     id   obs    pt rate       rate_l     rate_u
  <dbl> <dbl> <dbl> <chr>      <chr>      <chr>
1     1     0    10 "    0.000" "      NA" " 299.5732"
2     2    10    10 "1000.000" "507.9405" "1782.4887"
```

- First example: As the incidence density rate=0, the lower limit (rate_l) of 95%CI is set to missing by default.
- Second row: Incidence density rate is 1,000 per 1,000 person-time along with 95%CI.

Reference

1. Dean AG, Sullivan KM, Soe MM. OpenEpi: Open Source Epidemiologic Statistics for Public Health, Version 2.3.1. www.OpenEpi.com
2. Rothman KJ, Boice JD Jr: Epidemiologic analysis with a programmable calculator. NIH Pub No. 79-1649. Bethesda, MD: National Institutes of Health, 1979;31-32.
3. Rothman KJ, Greenland S. Modern Epidemiology, 2nd Edition. Lippincott-Raven Publishers, Philadelphia, 1998.

4. MID-P FOR COMPARING 2 RATES

Description

This function can be used to perform a comparison between two incidence density rates (e.g., CAUTI rates), consistent with the methods used by NHSN.

Example dataset and input parameters

Input parameters

O1= observed events in group-1

PT1=person-time in group-1
O2= observed events in group-2
PT2= person-time in group-2

```
#example dataset
df <- data.frame(
  id      = c(1,2,3,4),
  obs2017 = c(NA,1,1,1),
  pt2017  = c(2.3,2,2,2),
  obs2016 = c(2,3,3,0),
  pt2016  = c(4.5,-4,4,5)
)

# Apply the binom function row-wise
df1 <- df %>%
  rowwise() %>%
  mutate(binom_result = list(binom_rate(O1 = obs2017, PT1 = pt2017, O2 = obs2016, PT2 = pt2016))) %>%
  unnest_wider(binom_result)#%>%
  # mutate(across(c(midP, RATIO, LL, UL), ~ round(., 3)))    #format to 3 decimal places

# Display the results
head(df1)
```

Outputs and interpretations
Abbreviations
mid-P=significance test for comparing between 2 rates
RATIO = Rate in group-2 / Rate in group-1
LL=lower limit of 95%CI of rate ratio
UL=Upper limit of 95%CI of rate ratio
Note: When RATE_RATIO=0, the lower limit (LL) of 95%CI is set to missing by default. A user may change it to LL=0 if desired.

```
# A tibble: 4 x 9
     id obs2017 pt2017 obs2016 pt2016   midP RATIO     LL    UL
  <dbl>   <dbl>  <dbl>   <dbl>  <dbl>  <dbl> <dbl>  <dbl> <dbl>
1     1      NA    2.3       2    4.5 NA      NA    NA     NA
2     2       1    2         3   -4   NA      NA    NA     NA
3     3       1    2         3    4    0.790   1.5  0.160 39.5
4     4       1    2         0    5    0.286   0   NA      7.60
```

- First and second examples show missing ‘obs2017’ and negative value in ‘pt2016’ respectively so that the function would not generate any statistical output.
- As third and fourth example show midP >0.05, we would conclude that there is no significant difference between 2 incidence rates.

Reference

1. Erich L. Lehmann, Joseph P. Romano. Testing Statistical Hypotheses. Wiley, New York, 1959(Third edition, 2005, Springer Texts in Statistics)
2. John Pezzullo. Interactive Statistical Calculation Pages. www.Statpages.com.
3. Dean AG, Sullivan KM, Soe MM. OpenEpi: Open Source Epidemiologic Statistics for Public Health, Version. www.OpenEpi.com, updated 2013/03/21
4. Austin, Harland (Emory). Epidemiology method-II: Statistical issues for Density type follow-up studies.

5. MID-P FOR SINGLE PROPORTION

Description

This function can be used to perform confidence interval testing around a single proportion (e.g., healthcare worker flu vaccination percentages). This function can also be used to compare a single proportion to a benchmark value, consistent with methods used by NHSN

Example dataset and input parameters

Input parameters:

vx=numerator, vn=denominator.

```
# EXAMPLE DATAFRAME
df <- data.frame(
  id   = c(1, 2, 3),
  obs  = c(1, 10, NA),
  total= c(2, 4, 0.03)
)

# Apply the function row-wise and add multiple columns
df1 <- df %>%
  rowwise() %>%
  mutate(result = list(binom_midp_ci(obs,total)))%>%
  unnest_wider(result) #%>%
  mutate(across(c(vp, dl, du), ~ round(., 3)))    #format to 3 decimal places

# Display the results
head(df1)
```

Outputs and interpretations

Abbreviations

vp=proportion,

dl=lower limit of 95%CI,

du=upper limit of 95%CI

Note: When vp=0, the lower limit (dl) of 95%CI is set to missing by default. A user may change it to dl=0 if desired.

```
# A tibble: 3 x 6
     id   obs total    vp     dl     du
  <dbl> <dbl> <dbl> <dbl>  <dbl>  <dbl>
1     1     1  2      0.5  0.025  0.975
2     2    10  4     NA   NA     NA
3     3    NA  0.03  NA   NA     NA
```

- First example: Point estimate of single proportion of 1/2 is 0.5 with 95%CI provided.
- Second example has denominator (total) smaller than numerator so that there would be no statistical output.
- Third example has missing numerator (obs).

Reference

1. Dean AG, Sullivan KM, Soe MM. OpenEpi: Open Source Epidemiologic Statistics for Public Health, Version 2.3.1. www.OpenEpi.com
2. Rothman KJ, Boice JD Jr: Epidemiologic analysis with a programmable calculator. NIH Pub No. 79-1649. Bethesda, MD: National Institutes of Health, 1979;31-32.
3. Fleiss JL. Statistical Methods for Rates and Proportions, 2nd Ed. John Wiley & Sons, New York, 1981.
4. Newcombe RG. Two-sided confidence intervals for the single proportion: comparison of seven methods. Statistics in Medicine 1988;17:857-872.
5. Stein Vollset. Confidence intervals for a binomial proportion statistics in medicine,12,809-824 (1993)

6. MID-P FOR COMPARING 2 PROPORTIONS

Description

This function can be used to perform a comparison of two proportions (2 x 2 contingency table), ie, to compare the risks (prevalence) of outcome in two independent groups.

Example dataset and input parameters

Input parameters:

- exposed with disease (A),
- not-exposed with disease (B),
- exposed without disease (C),
- not-exposed without disease (D).

Letters refer to positions in classic 2X2 exposure/disease status table.

```
Data input layout:
A= E+,D+
B= E-,D+
C= E+,D-
D= E-,D-
                                  Disease/Outcome
                          |    Yes                 No            |
                -----------|-------------------------------------|-------
        Exposure     Yes  |     A                   C            |
                     No   |     B                   D            |
                -----------|-------------------------------------|-------
```

```
# Example dataset
df <- data.frame(
  id= c(1,   2, 3,  4),
  A = c(10, 20, 3, -3),
  B = c(20, 60, 5,  5),
  C = c(11, 80, 7, NA),
  D = c(21, 100, NA, 7)
)

# Apply the function row by row using dplyr
df1 <- df %>%
  rowwise() %>%
  mutate(result = list(twobytwo(A, B, C, D))) %>%
  unnest_wider(result)#%>%
  #mutate(across(c(Mid_P), ~ round(., 3)))    #format to 3 decimal places

# Display the results
head(df1)
```

Outputs and interpretations

```
# A tibble: 4 × 6
     id     A     B     C     D    Mid_P
  <dbl> <dbl> <dbl> <dbl> <dbl>    <dbl>
1     1    10    20    11    21  0.934
2     2    20    60    80   100  0.00275
3     3     3     5     7    NA  NA
4     4    -3     5    NA     7  NA
```

- First example shows mid-p=0.00275 indicating that proportion (prevalence, risk) of disease in exposed group (A/(A+C)) is not statistically different from that of non-exposed group (B/(B+D)).
- Mid-p=0.934 in second example indicates that proportion (prevalence, risk) of disease in exposed group is statistically NOT different from that of non-exposed group.
- Other examples have missing cell value or incorrect value such as negative value, mid-P is not computed.

Reference

1. Dean AG, Sullivan KM, Soe MM. OpenEpi: Open Source Epidemiologic Statistics for Public Health, Version 2.3.1. www.OpenEpi.com

2. Rothman KJ, Boice JD Jr: Epidemiologic analysis with a programmable calculator. NIH Pub No. 79-1649. Bethesda, MD: National Institutes of Health, 1979;31-32.

3. Fleiss JL. Statistical Methods for Rates and Proportions, 2nd Ed. John Wiley & Sons, New York, 1981

7. Comparing Two ECDFs

Description

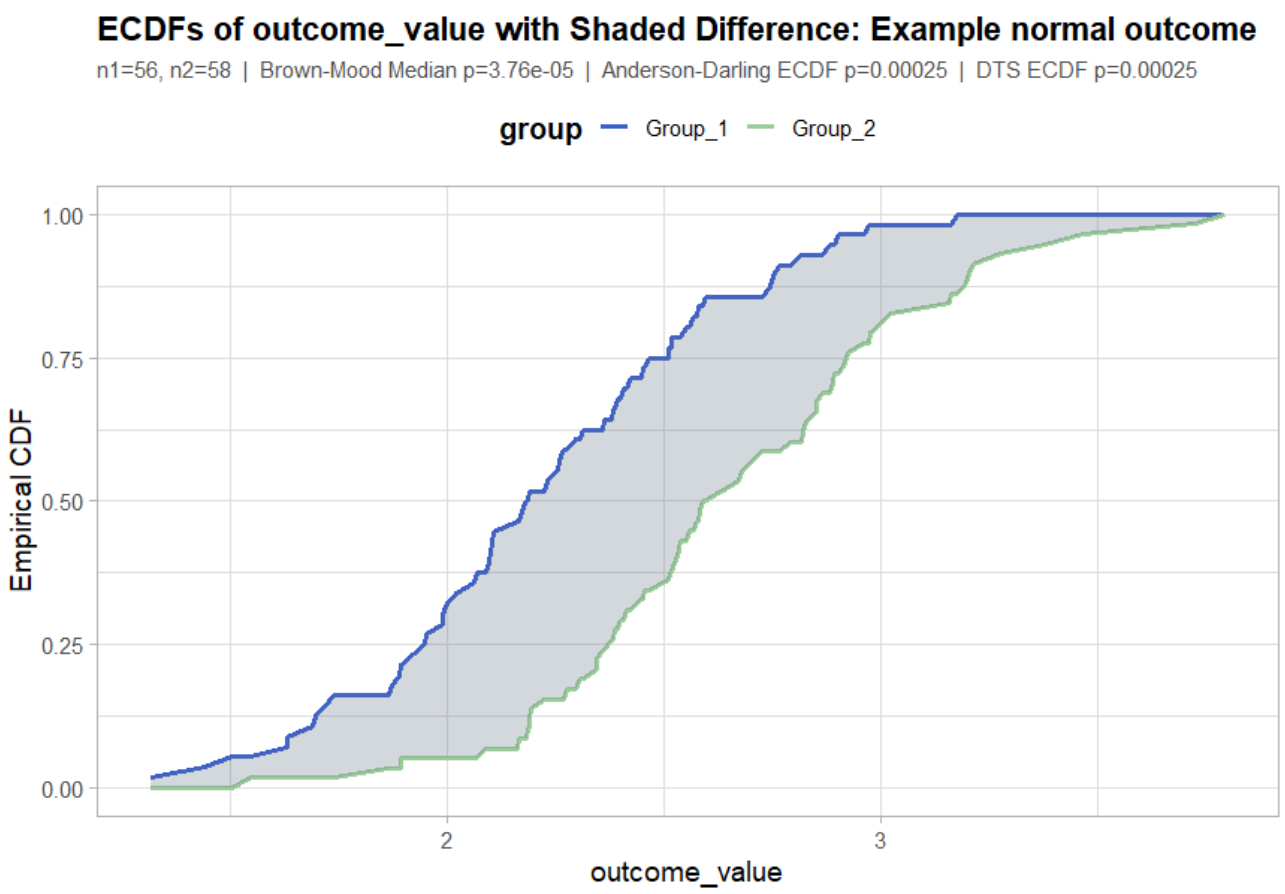
This R function compares medians and ECDFs of a single numeric outcome for two samples or groups using Brown-Mood median and both Anderson-Darling and Dowd DTS ECDFs tests.

Example data and Input parameters

```
#-----------------------------------------------------------------------------#
#   Example:  Generic data frame from a normal distribution                   #
#             function: compare_two_ecdfs                                     #
#                                                                             #
#-----------------------------------------------------------------------------#

set.seed(123)

example_df <- data.frame(
  subject_id = 1:120,
  group = rep(c("Group_1", "Group_2"), each = 60),
  outcome_value = c(
    rnorm(60, mean = 2.2, sd = 0.45),   # Group 1 distribution
    rnorm(60, mean = 2.7, sd = 0.50)    # Group 2 distribution
  )
)

# Introduce a few missing values (optional, realistic)
example_df$outcome_value[sample(1:120, 6)] <- NA

str(example_df)


compare_two_ecdfs(
  data = example_df,
  outcome = "outcome_value",
  group_var = "group",
  group1 = "Group_1",
  group2 = "Group_2",
  plot_label = "Example normal outcome"
)
```

Output:

The main output is a plot of the two ECDFs with shaded area in between regardless of whether they cross each other along the range of values of the outcome variable called “outcome_value.” This plot is further annotated with data in the subtitle.



Interpretation:
The title references the name of the outcome being compare called “Example normal outcome” and the subtitle include the number of observations for each sample and p-values for comparing the two medians and comparing two ECDFs using two different tests. In this example, there are 56 observations in sample 1 and 58 observations in sample 2. The Brown-Mood median p-value is displayed using scientific notation as 3.76e-05. This is equivalent to p=0.0000376 and is sufficiently small (<0.05) and statistically significant. The p-values for the two ECDF tests, namely, Anderson-Darling and DTS are identical at p=0.00025 yielding a statistically significant result when comparing ECDFs.

REFERENCE:

1. **Brown, G. W., & Mood, A. M. (1951).** On Median Tests for Linear Hypotheses. *Proceedings of the Second Berkeley Symposium on Mathematical Statistics and Probability*, 159–166. University of California Press.

2. **Anderson, T. W., & Darling, D. A.** (1952). "Asymptotic theory of certain 'goodness-of-fit' criteria based on stochastic processes." *Annals of Mathematical Statistics*.
3. **Dowd, C.** (2020). *A New ECDF Two-Sample Test Statistic*. arXiv preprint arXiv:2007.01360. Available at https://arxiv.org/abs/2007.01360 or https://codowd.com/public/DTS.pdf